Document ID: VIB-TECH-API-001 Classification: INTERNAL CONFIDENTIAL Effective Date: 2024-04-01 Version: 1.2

Document Owner: Head of Enterprise API Strategy & Integration

API Catalog: VI Business (India)

1. Executive Purpose and Document Scope

This API Catalog serves as the definitive reference for the enterprise application programming interfaces (APIs) exposed by VI Business (India). It is designed to guide enterprise customers, third-party system integrators (SIs), and internal solution architects through the integration of VI Business managed services, network provisioning, and billing capabilities into their native IT service management (ITSM), enterprise resource planning (ERP), and monitoring toolsets.

This document is prioritized for enterprise proposal responses, technical bid defenses, and implementation planning. It ensures consistent messaging regarding our integration capabilities and boundaries.

2. Definitions, Assumptions, and Applicability

Definitions:

- **API Gateway:** The centralized, load-balanced Kong-based gateway hosted in our primary Tier-III Mumbai (NESCO) core that proxies all inbound client requests.
- **cAPN:** Custom Access Point Name, used for private IoT and enterprise mobility routing.
- **TPS:** Transactions Per Second.

Assumptions:

- Enterprise customers must possess a valid Master Services Agreement (MSA) and a dedicated VI Business Account Manager before receiving production API credentials.
- Data residency regulations adhere rigorously to the DPDP Act 2023; all API endpoints resolve to instances physically hosted within Indian borders.
- External developers are required to use TLS 1.2 or higher for all transmission.

Applicability:

- **Sales & Pre-Sales:** Use as a factual artifact for RFP responses questioning integration capabilities.
- **Customer Success:** Guide onboarding technical workshops.

3. Authentication Model and Security

All VI Business APIs utilize a standard, robust security model designed to meet banking, financial services, and insurance (BFSI) sector requirements.

3.1 Authentication Mechanism

We enforce **OAuth 2.0 (Client Credentials Grant)** for all machine-to-machine (M2M) API interactions.

1. Customers obtain a `client_id` and `client_secret` via the VI Business Enterprise Self-Care Portal after multi-factor authentication (MFA) approval by the authorized customer admin.

2. The client application calls the `/oauth2/v1/token` endpoint to receive a short-lived JSON Web Token (JWT).
3. The JWT must be presented in the authorization header as a Bearer token: `Authorization: Bearer <token>`

3.2 Security Posture

- **Encryption in Transit:** Strict enforcement of TLS 1.2 and TLS 1.3.
- **IP Whitelisting:** Optional but highly recommended restricted access control requiring source IP CIDR registration.
- **Data Masking:** PII elements returned in payloads (e.g., end-user names on enterprise mobile plans) are partially masked based on the assigned API role.

4. Available APIs (Core Catalog)

The active enterprise API catalog is divided into functional domains mapping to our managed service portfolios.

4.1 Enterprise Mobility and Managed IoT

API Name	Endpoint Path	Description	Key Use Case
SIM Lifecycle Management	`/iot/v2/sim/lifecycle`	Activate, suspend, or terminate SIM/eSIM profiles on the VI network.	Custom ERP integration for automated POS terminal activation.
Data Usage Analytics	`/iot/v2/usage/metrics`	Retrieve real-time aggregate and per-device granular data consumption metrics.	Fraud detection and automated threshold alerting for connected fleets.
Location Tracking (LBS)	`/iot/v2/location/cell-id`	Securely expose the real-time Cell-ID location of an enterprise mobility asset.	Logistics tracking for high-value asset transport within India.

4.2 Managed Network Services (SD-WAN & MPLS)

API Name	Endpoint Path	Description	Key Use Case
WAN Performance Telemetry	`/network/sdwan/v1/telemetry`	Extrapolate latency, jitter, and packet loss metrics across primary (Optical) and secondary (LTE/5G) links.	Single-pane-of-glass integration into a customer's native ServiceNow instance.
Bandwidth on Demand (BoD)	`/network/mpls/v1/bod`	Dynamically scale bandwidth for designated MPLS ports (up to 2x commit) on an hourly billing model.	Addressing seasonal traffic spikes for e-commerce retailers across Tier-1 Hubs.

4.3 Billing and Service Assurance

API Name	Endpoint Path	Description	Key Use Case
Consolidated Invoicing	`/billing/v1/invoices`	Retrieve unbilled usage, download PDF invoices, and review line-item charge granularity.	Automated financial reconciliation process in SAP or Oracle ERP.
	`/support/v1/incidents`		

API Name	Endpoint Path	Description	Key Use Case
Ticketing Integration		Bidirectional synchronization for creating, updating, and querying P1-P4 incidents handled by the VI NOC/SOC.	Bridging enterprise ITSM tools directly to VI Support, eliminating manual email logging.

5. Payload and Integration Patterns

- **Format:** All APIs conform to the RESTful architecture paradigm.
- **Data Serialization:** Payloads utilize strict JSON (`application/json`) format. XML support was deprecated in Q4 2023.
- **Pagination:** Endpoints returning lists leverage standard limit/offset pagination, with a default `limit` of 50 and maximum `limit` of 500 records per response.
- **Idempotency:** Specific state-changing POST requests (e.g., SIM activation) accept an `Idempotency-Key` header to prevent duplicate billing via accidental retries.

6. Versioning Strategy

- **URI Versioning:** The major version is embedded directly in the URL path (e.g., `/v1/` , `/v2/`).
- **Backward Compatibility:** Additions of new JSON fields are considered backward-compatible. Breaking changes (e.g., removing a field, changing data types) require a major version increment.
- **Deprecation Window:** A minimum of 12 months notice is provided to registered API technical contacts before a major API version is forcibly sunset.

7. Rate Limits and Quotation

To preserve gateway stability and ensure 99.99% availability for all enterprise tenants, strict rate limiting is applied universally, segmented by account tier.

Endpoint Category	Standard Limit	Burst Allowance	Limit Reset Window
Authentication	10 TPS	15 TPS	1 second
Telemetry/Reporting	50 TPS	100 TPS	1 second
Lifecycle/Provisioning	5 TPS	10 TPS	1 second

Note: Custom rate limit elevations (+500 TPS) are available exclusively under dedicated Commercial Priority Support contracts, subject to VI Core Networks capacity approval.

8. Error Handling

VI APIs adopt standard HTTP status codes combined with a detailed JSON error payload schema to facilitate programmatic debugging.

Standard HTTP Codes:

- `200/201/202` : Success variants.
- `400 Bad Request` : Validation failure.
- `401 Unauthorized` : Missing or expired JWT.
- `403 Forbidden` : Insufficient scopes/roles for the requested action.
- `429 Too Many Requests` : Rate limit exceeded.
- `500/502/503` : VI Internal Gateway or Backend errors.

Error Payload Standard Structure:

```
{
  "error": {
    "code": "VALIDATION_FAILED",
    "message": "The 'iccid' provided does not meet the 19-digit length requirement.",
    "tracking_id": "req_8a3bd472ecc14d"
  }
}
```

9. Event/Webhook Support

Instead of forcing clients to persistently query for status changes, VI Business offers a robust asynchronous Webhook architecture.

- **Supported Events:** Incident status changes, Invoice generation, SIM lifecycle state transitions, Data usage threshold alerts.
- **Delivery Mechanism:** HTTPS POST requests to registered customer callback URLs.
- **Security Validation:** Webhook payloads are cryptographically signed using a unique symmetric key shared during webhook registration. Clients must validate the `X-VI-Signature` HMAC-SHA256 header to authenticate the event's origin.
- **Retry Policy:** If the destination returns a non-2xx status, VI will attempt exponential backoff retries for up to 24 hours.

10. Constraints, Exclusions, and Dependencies

- API availability is strictly subject to the underlying platform maintenance windows, which are scheduled between 02:00 IST – 04:00 IST on Sunday mornings.
- Legacy physical copper-line services are not supported under the newer REST API umbrella.

11. Proposal-Use Guidance

- **RFP Mapping:** When prompted for "ITSM Integration" or "Automated Provisioning," explicitly reference this document. Do not invent endpoints.
- **Verification:** For security questionnaires, confirm OAuth 2.0 and TLS 1.2+ encryption unconditionally.
- **Custom Integration:** If an RFP demands custom ERP integration not natively covered by these endpoints, escalate to the Solution Architect team for professional services scoping.

12. Governance and Ownership

- **Document Owner:** Head of Enterprise API Strategy & Integration
- **Reviewers:** IT Delivery Head, Enterprise CISO (Rajesh Kumar), Enterprise Deal Desk
- **Update Cadence:** Bi-annually or aligning with major API product release cycles.